“Transit Equity in the Face of the COVID-19 Pandemic: A Study of Six Ontario Transit Authorities”

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Abstract

The COVID-19 pandemic has created severe challenges for public transit systems. Ridership has dropped dramatically and remains below pre-pandemic levels for three main reasons: public fears regarding the safety of public transit, the number of people now working from home, and the increased use of alternative means of transportation. Those most reliant on public transportation have been most negatively affected by cuts to public transit made in response to the pandemic.

This paper examines the pandemic's impact on six Ontario transit systems. Using qualitative data from interviews of public transit officials and municipal leaders and secondary data from public documents and news reports, the paper examines whether measures introduced by the transit authorities addressed the issue of transit equity and the extent to which vulnerable groups were included in transit planning in the first year of the pandemic. The findings are that transit officials were aware of transit equity and did know that transit service cuts disproportionately affected the most vulnerable. They also proposed measures to mitigate the impact of cuts on vulnerable groups and raised transit equity in their appeals to senior government for more funding. They were, however, limited in their ability to address transit equity. The transit systems were forced to prioritize the maintenance of existing levels of service during the pandemic with only a limited capacity to introduce measures to advance transit equity.

Keywords: public transit; transit equity; pandemic; Ontario; transit poverty

1. Introduction

The COVID-19 pandemic has created a crisis for public transit officials and users. As the world was shutting down in an attempt to halt the spread of the pandemic, so too was there a collapse in the use of public transit in almost every jurisdiction. Various public transportation systems in North America saw drops in their ridership of up to 90% (Ahangari et al., 2020: 3; Dachis & Godin, 2021). In Toronto, for example, the Toronto Transit Commission (TTC) – the largest transit system in Canada – reported that ridership was around 15% of pre-pandemic norms approximately one month after the scale of the pandemic became apparent, and that ridership had only returned to around 30% some six months later. As of April 2021, ridership revenue from the TTC was only 36% of the pre-pandemic levels (Chief Executive Officer - TTC, 2021: 5). Smaller transit authorities throughout Ontario witnessed similar ridership declines with most experiencing an immediate drop of between 80 and 90% followed by a slow increase that still saw ridership at between 40 and 50% of pre-pandemic levels over a year after the onset of the pandemic.

Across North America, the COVID-19 pandemic continues to have a major impact on public transit. Transit decision-makers face a situation where transit use has declined substantially, and costs have increased – as a result of implementing new cleaning procedures, for example – at the same time as revenues have declined. Starting from the realization that the COVID-19 pandemic has had a major impact on public transit, this paper examines the measures introduced by transit authorities in response to the pandemic. Of particular interest is the impact of these measures on individuals and groups that rely on public transit and the extent to which decision-making has addressed transit equity. While the definition of transit equity is contested, it broadly refers to the provision of transit services for those most reliant on public transit with a focus on the level of service provided, the affordability of service, and the accessibility of that service (see Allen & Farber, 2019; Amar & Teelucksingh, 2015; Wray, 2013; Sengupta et al., 2013). Various studies identify that certain groups – newly immigrated individuals and families, the elderly, and low-income community members – are most likely to rely primarily on public transit to get to work, social services, or to shops but are most likely to suffer from inadequate transit service. In addition, a lack of transit equity may also be evident in the context of transit planning and decision-making with disadvantaged groups unable to participate effectively in transit decision-making processes, even in contexts when public participation is encouraged or required (Aubin & Bornstein, 2012; McAndrews & Marcus, 2015). This lack of access to public transit, or alternative transit options, reflects existing inequalities in society and in turn perpetuates these inequalities.

Given the pandemic's negative impact on public transit systems, and given that this disproportionately affects those who continue to rely on public transit, it is important to examine whether transit authorities have taken transit equity into account in their pandemic decision-making and whether the measures implemented address transit equity.

2. Methodology

This paper addresses three main sets of questions relating to transit equity. First, to what extent has the pandemic affected public transit services? Second, to what extent were transit users consulted in the pandemic decision-making process and if so, did this include marginalized groups? And third, to what extent did transit decision-makers take transit equity into account in their decision-making? These questions are examined using a comparative case study of public transit in six Ontario transit authority areas: Brampton, Brantford, Burlington, Kingston, London, and Windsor. These were used as comparative cases because they have public transit systems that rely primarily on buses. As they are also in the same provincial jurisdiction, they have been affected by the same provincial government decisions and funding during the pandemic.

Table 1: Transit Authority Data - 2019

Transit Authority	**Municipality size**	**Number of Buses**	**2019 Ridership**
Brampton	593,638	450	31,914,291
Brantford	98,179	33	1,905,338
Burlington	193,668	63	2,450, 395
Kingston	123,069	79	6,929,280
London	346,765	220	24,599,655
Windsor	217,188	114	8,430,750

The transit authority areas range in size from a low of approximately 100,000 population and 33 buses (Brantford) to a high of approximately 600,000 and 450 buses (Brampton). The transit authorities therefore differ in size and operating budgets. This study of similar but different transit authorities allows for an examination of whether the responses to the pandemic are similar, or whether they are unique to each city or affected by the size of the transit system.

Data for this project were drawn from semi-structured interviews with eleven transit decision-makers in the six authorities. These semi-structured interviews were conducted over video link in December 2020; that is, some nine months into the pandemic. As Verbich et al. identify, questioning transit agency representatives on the objectives they use and consider significant provides important information about the prevalence of transit equity in transit planning (2017: 60). Interviewees were asked a series of questions (and follow up questions) relating to public transit during the pandemic, their authority's response to the pandemic, what they see as the major problems facing transit moving forward, and their views on how public transit can survive the pandemic. Interviewees were not asked specifically about transit equity until near the end of their interview. This interview strategy was used to determine whether transit equity – or associated concepts such as transit poverty, disadvantaged groups – were mentioned by the interviewees without prompting. Specific questions about transit equity were asked at the end of the interview or as follow up questions in cases where the interviewee specifically raised the concept.

Prior to analyzing the data provided by the interviewees, interviewees were given the opportunity to edit their transcripts and add additional information if needed. These semi-structured interviews provided qualitative data directly from those involved in public transit decision-making within the six cities. Interviewees were able to provide us with city-specific data and personal experiences and information related to public transit in the COVID-era. Secondary data was used to supplement the information collected in the interviews. This secondary data included municipal websites, municipal publications, and announcements, as well as academic sources, think tanks, and news sources. In examining these secondary sources, we looked particularly for information about pandemic-induced changes to transit funding, transit services, and public consultation during the pandemic.[1]

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[1] Our interviews were limited to public transit decision-makers. With the benefit of more time and resources, surveys and interviews of riders and bus operators would have allowed for a more comprehensive look at public

3. Theory

3.1 Transit Equity

Transit equity is a much-studied concept, and, as with studies of equity in other contexts, there is disagreement about its definition, causes, measurement, and what can be done to promote it. At its core, transit equity relates to access to transit service within the wider transportation system and the fairness of that system in terms of the costs and benefits to those using and paying for the system (Campbell, 2019; Litman, 2002). It is an awareness that some people are disadvantaged and that this is reflected in the existing transportation system and indeed may be caused by transit decision-making (Welch & Mishra, 2013). Promoting greater transit equity is about recognizing these inequalities and designing a system to overcome them.

Scholars identify two types of equity to help understand transit equity (Carleton & Porter, 2018). They define *horizontal equity* as equally distributing costs and benefits among people within an overall population so that residents with similar needs and resources are treated equally by the transportation system (Allen & Farber, 2019). This can be connected to geographic location and the availability of transit options no matter where residents are located within the transportation system. It is also connected to the costs of different transit options and whether these costs are distributed equally relative to what individuals pay and their ability to pay. A focus on horizontal equity, for example, may identify that the wider transportation system subsidises car drivers – by not charging for the full cost of road construction or parking – over users of bus services, or that commuter rail trips are subsidised more per ride than buses services.

transit and the impact of COVID-19 on the service. We are also aware that the pandemic is continuously evolving and a number of transit-related developments have occurred since the completion of our interviews.

Vertical equity, on the other hand, is concerned with the distribution of transit services among different groups of users (Allen & Farber, 2019; El-Geneidy et al., 2016; Iseki, 2016: 2). In this view, different groups have different needs with respect to the transportation system and currently have different access to transit options and experience different levels of service. Equity is advanced if transit services are provided in a way that helps disadvantaged groups, through subsidized service or if more transit options are provided to users lacking access to private transportation. The latter group may, for example, include seniors or people with disabilities who cannot drive and low-income residents who cannot afford to own and operate a private automobile.

These definitions of transit equity overlap. Underlying them both is the argument that certain residents and/or groups of users enjoy better and more varied transportation options than others. Examinations of existing transportation systems certainly point to the existence of inequality and transit poverty in these systems. It is, for example, frequently noted that certain categories of residents – including seniors, students, recent immigrants, and low-income individuals – rely heavily on public transit and that the quality and quantity of this service are frequently problematic (Byckalo-Khan & Gosine, 2003; Sengupta et al., 2013: 6). In many urban areas, for example, transit services geared toward commuters to and from city centres, business centres and entertainment locations are prioritized at the expense of services within other, poorer, parts of the urban area. The re-introduction of streetcar systems in several North American cities is an example of a transit service that often primarily meets the needs of wealthier residents and tourists (Culver, 2017; Lowe & Grengs, 2020).

Studies of transit systems across North America find that lower income residents are not well served by those systems and problems relating to the affordability, availability, frequency, and accessibility of transit options can lead to the exclusion of marginalized groups in society (Wray, 2013: 5). Insufficient transit options – and resultant long commute times – can, for example, act as a barrier to both employment and other necessary services (Allen & Farber, 2019; Amar & Teelucksingh, 2015: 55; Hertel et al, 2016; Manaugh et al, 2015; Sengupta, Fordham & Day, 2013). Even where public transit service is available, it is possible that its cost limits the ability of those on low income to access the service (Sengupta et al, 2013: 19; Verbich & El-Geneidy, 2016; Griffin & Sener, 2016: 133). In many cases the individuals that need public transit the most are those most likely to be trapped in geographic or economic circumstances that provide them with the lowest accessibility to public transit (Wray, 2013: 29). This is sometimes connected to the concept of *spatial mismatch*; a situation where low-income workers live far from locations where there are places of employment (Bauder, 2000; Ong and Miller, 2005). Grengs argues that this is sometimes better referred to as a *modal mismatch*, a situation whereby residents are separated from jobs because they lack a private automobile and public transit is either unavailable or inadequate (2010). Allen and Farber have coined the term *transport poverty* whereby the combination of socioeconomic disadvantages and lack of access to transport services leads to limited access to important destinations for low-income communities (2019).

Whatever term is used and however the situation is defined, there is a clear connection between marginalization and lack of access to public transit in Canada (Amar & Teelucksingh, 2015; Wray, 2013). This marginalization is not exclusively based on poverty – or rather poverty often intersects with other forms of disadvantage. In a Toronto study, for example, single mothers, those living on low-incomes, and newcomers are found to be the groups most impacted by inaccessible transportation services, which in turn impacts other aspects of their lives including (but not limited to) access to food, access to recreation and access to health care

(Wray, 2013: 4). Another study in Quebec found that low-income women have limited daily activity in the community due to inaccessible public transit (Allen & Farber, 2019: 5). Various studies identify that immigrants are frequently disadvantaged by existing transportation systems. In Toronto, for example, nearly 36.3% of immigrants use public transit and in particular women who have immigrated to Canada are heavily reliant on public transit and have fewer options to get to work (Amar & Teelucksingh, 2015). Often immigrants cannot afford the higher housing costs connected with living transit-rich neighbourhoods and are forced to live in neighbourhoods with lower levels of public transit service and are thus negatively affected economically and socially (Carleton & Porter, 2018: 65; Karner, 2012: 46).

There is, moreover, a connection between rising rates of citizens experiencing homelessness and the use of public transit, often for shelter. As housing affordability continues to decline and there remains limited shelter availability, those experiencing homelessness often turn to buses and bus stops for a form of shelter or to use it to get to social services (Ding et al, 2021).

3.2 Promoting Transit Equity

For many individuals, then, public transit is an essential service that allows them to get to important destinations including but not limited to their workplace. It is, therefore, vital that transit plans focus those who do not have other options for transportation and who need public transit services. Transit authorities need to address the many factors that impact accessibility to public transit including distance from bus stops, frequency of buses, and time spent on buses (Foth et al., 2013: 2). In addition, planning must also address underlying factors that impact equity and the accessibility of transit services, such as fare prices and fare options (El-Geneidy et al., 2016: 4).

Promoting transit equity is dependent on at least three factors. First, and most significantly, is resources. No transit system is entirely self-sufficient with the cost of offering the service covered by user fees. Prior to the pandemic, for example, approximately 50% of Transit Windsor's expenses were covered by fares with the rest depending on taxpayer dollars. Kingston and Burlington were similar with only 35 to 40% of expenses covered by transit fares prior to COVID-19. Even before the pandemic, then, public transit depended on financial support from municipal, provincial and federal governments. Resources are finite and decisions about funding dedicated to public transit are made by governments that must balance these expenditures against other budgetary demands. In the absence of unlimited funding, transit planners, as well as the governments responsible for overseeing these transit authorities, need to establish priorities and make decisions about levels of service, service coverage, transit affordability, and its financial viability. All of these decisions have the potential to affect transit equity positively or negatively (Griffin & Sener, 2016: 136; Verbich et al., 2017). An increase in the transit budget may, for example, allow a transit authority to expand or increase services to underserviced communities. A decision to build a new subway or light rail line will likely reduce the budget available to expand existing bus routes. Even with the best of intentions, it is not possible for transit planners to provide the types and levels of service demanded by all users and all advocates of transit equity.

Transit planning, then, involves making choices. This draws attention to the fact that transit planning affects transit equity. Transit planning can positively or negatively affect transit equity depending upon who is listened to and what factors are taken into account during the planning process in establishing the transit system's priorities. Achieving transit equity requires attention to transit planning through an inclusive and accessible public participation process (Hertel et al., 2016; Karner, 2012: 59). The key questions here concern the extent to which transit users are allowed to participate in the planning process. If transit users do participate, it is also important to ask whether the participants are representative of marginalized communities and whether decision-makers incorporate this participation into transit plans (Sutcliffe & Cipkar, 2017). Studies of citizen engagement in politics in general, and transit planning in particular, frequently raise critical questions about the extent to which public participation is representative and effective (Fung, 2006; McAndrews & Marcus, 2015; Martens et al, 2012).

A third factor that affects transit equity, and which relates to the previous two, concerns the measurement of transit equity. Public transit systems may claim to strive towards transit equity but determining whether they are successful is complicated. A 2015 study of transit services in North America, for example, found that while many public transit plans claim to advance social equity, most do not have clear measures for determining whether they are successful in doing so (Manaugh et al., 2015). Others argue that there is currently no agreed upon framework to measure transit equity which has led most studies to focus on qualitative methods to review transit equity in practice (Xiaohong, Jihao & Peng, 2020). Without clearly specified objectives and measures, it is difficult to measure social equity impacts and in turn, difficult to prevent social inequities from persisting.

4. Results and Discussion

4.1 The Impact of the Pandemic

Transit ridership collapsed as the pandemic took hold and as governments around the world decided to lock down by restricting movement and trying to limit work-related travel to essential workers only with others forced to work for home. Many who did have to continue commuting to work found alternative travel means either because they were forced to do so as a result of reduced public transit service levels or so as to avoid health risks associated with using public transit.

The impact on transit ridership was immediate and dramatic. In their annual review of transportation, Transport Canada spelled out the extent of the decline in the use of public transit.

> There was a drastic reduction in public transportation ridership throughout 2020. In 2020, public transit systems carried around 849 million passengers, a 55% decrease from 2019. During the early stages of the pandemic the loss of ridership was much larger, reaching upwards of 90%. The sector began a period of recovery during the summer months, when daily COVID-19 cases were low, with consecutive months of growth from June-September. Following this rebound, ridership declined again as the country entered the second wave of cases. December marked the tenth month of decline year-over-year, with networks carrying 52.5 million passengers, down 65.8% from December 2019 (Transport Canada, 2020: 87).

The transit systems in this study witnessed significant immediate declines in public transit use (see Table 2). The situation in Windsor stands out as a result of the mayor's March 2020 decision to suspend all public transit. During this period, Transit Windsor carried no passengers. In the first month following the resumption of public transit in Windsor, ridership was approximately 90% lower than in the last full month prior to the lock down; a figure that was in line with the other five transit authorities. All of the transit authorities carried significantly fewer passengers in 2020 than they had done in 2019. Burlington Transit, for example, saw its total ridership shrink from 2.45 million in 2019 to 1.47 million in 2020. Brampton's total 2020 ridership was only 43.29% of its 2019 total, and London Transit's ridership declined by over 48% in the same period. The financial impact on the transit authorities was even greater than could be surmised from the reduction in ridership alone. To protect their drivers each of the authorities introduced rear-door boarding at the onset of the pandemic, which also meant that they were unable to collect fares. They were only able to resume collecting fares once plexiglass screens had been installed (see below).

While transit usage in these systems has increased as the pandemic moves into its second year, and all transit authorities have resumed fare collection, ridership numbers remain lower than prior to the pandemic.

Table 2: Ridership Rates During the Pandemic

	Brampton			Brantford		
	2020	2021		2020	2021	
Month			**Change**			**Change**
Jan.	2,822,536	1,084,059	-62%	170,461	83,812	-50.8%
Feb.	2,544,244	1,018,409	-60%	160,729	72,772	-54.7%
Mar.	1,900,000	1,354,344	-29%	138,848	72,989	-47.4%
Apr.	640,000	1,236,782	+93%	49,611	66,661	+34.4%
May.	790,000	1,382,352	+75%	55,194	66,326	+20.2%
June	990,000	1,557,357	+57%	70,657	72,494	+2.6%
July	1,292,093	1,679,044	+30%	77,210	92,262	+19.5%
August	1,423,229	1,804,426	+27%	67,283	96,759	+43.8%
Sept.	1,543,710	2,045,907	+33%	88,258		
Oct.	1,522,300			95,124		
Nov.	1,420,870			92,432		
Dec.	1,209,254			63,614		
	18,098,238 -43.29%					

	Burlington			Kingston		
	2020	2021		2020*	2021	
Month			**Change**			**Change**
Jan.	224,223	88,351	-61%	649,064	117,939	-82%
Feb.	206,461	91,030	-56%	571,929	156,857	-73%
Mar.	145,612	128,173	-12%	372,565	227,643	-39%
Apr.	54,763	99,416	+82%	229,043	142,852	-38%
May.	68,785	99,985	+45%	203,123	139,391	-31%
June	91,090	120,030	+32%	190,893	168,388	-12%
July	101,996	142,506	+40%	190,591	198,187	+4%
August	98,760	142,175	+44%	186,866	210,308	+12.5%
Sept.	128,800	164,760	+28%	207,071	332,093	+60%
Oct.	132,303	169,559	+28%	212,844		
Nov.	129,014			212,112		
Dec.	118,151			181,162		

	London			Windsor		
	2020	2021		2020	2021	
Month			**Change**			**Change**
Jan.	2,347,400	387,506	-83%	826,680	108,289	-87%
Feb.	2,173,151	443,466	-80%	770,487	121,935	-84%
Mar.	1,835,877	653,258	-64%	512,239	179,846	-65%
Apr.	536,296	446,926	-17%	-	130,305	-
May.	631,340	440,649	-30%	72,266	127,559	+77%
June	828,251	510,558	-38%	141,128	146,233	+4%
July	970,300	582,352	-67%	157,330	171,431	+9%
August	786,749	619,454	-21%	189,714	181,283	-4%
Sept.	669,625	1,051,685	+60%	272,665	313,381	+15%
Oct.	661,663	1,113,619	+68%	203,608	-	
Nov.	671,587			171,538	-	
Dec.	568,479			145,975	-	

* The Kingston figures for months of April through August 2020 inclusive are City of Kingston estimates.

All of the interviewees in this study were aware that the longer-term consequences of the pandemic are potentially serious for the future of public transit. These can be broken down into three interrelated concerns. First, ridership may never return to pre-pandemic levels if people continue to work from home and education transitions to more digital learning in the post-pandemic era. Second, if services are cut and service quality declines, people who can, will find alternative transportation solutions. Several interviewees pointed to an increase in the number of automobile sales during the pandemic and noted that if people find reliable alternatives, it is difficult to attract them back to public transit. Third, interviewees noted that the pandemic has, to some extent, increased the pre-existing negative stigma that is often attached to public transit use, with public transit increasingly likely to be seen as a service only used by those without the means to buy a car.

In sum, the pandemic has had a massive impact on public transit evident primarily in reduced ridership. Although ridership has increased from the initial pandemic-collapse, decision-makers are aware that reduced ridership has the potential to have a lasting, negative effect on public transit. All of the interviewees either directly or indirectly identified the fear that public transit could enter a 'death spiral' as a result of the pandemic with lower ridership leading to lower revenue and lower service provision (such as reduced bus frequencies), leading to yet lower ridership and more pressure to reduce service (see also Hart, 2020; D'Angelo, 2020). The interviewees were uniformly aware that reduced service provision, and the threat of a death spiral, affected disadvantaged riders the most (see below). Several interviewees, for example, noted that working from home during the pandemic has been a privilege of high-income earners, whereas those in low-income groups remained reliant on public transit (see also Tirachini & Cat, 2020: 7). The interviewees were all aware that low-income earners were most affected by reductions in service provision

4.2 The Role of Transit Equity in Pandemic Planning

Transit decision-makers were forced to respond immediately to the new reality of transportation during the pandemic. Before turning to the decisions taken by transit authorities, it is important to examine the extent to which decision-makers consulted with interested parties and, particularly for the purposes of this paper, marginalized groups. Any examination of transit equity, as noted earlier, must assess whether the planning process engages with marginalized groups and takes their input into account.

The unexpected nature of the pandemic meant that transit authorities were forced to make decisions quickly with incomplete or limited information and little time to engage in consultations with transit users. A clear example of this was Windsor Mayor Drew Dilkens' decision to shut down all Transit Windsor services at the outset of the pandemic. The decision was made unilaterally using the Emergency Measures Bylaw that gives the mayor executive authority over all municipal matters (Cross, 2021a). One interviewee suggested that the mayor had the best intention in mind when making this decision and identified several factors that put Windsor at a higher risk – such as the extensive connections with Detroit where the incidence of COVID was higher – that influenced his decision. The majority opinion, however, was that this decision was a mistake and one that removed an essential service from Windsor for four weeks and left those reliant on public transit excluded from society. As one interviewee noted, nearly 2,000 to 3,000 riders were left scrambling to get to work or make other essential trips. As the interviewee identified, "a trip that may have cost them $4 prior to the pandemic could end up costing $30 in taxi fares."

Transit groups and users were not consulted prior to this decision being taken and indeed the decision sparked considerable opposition. Windsor residents spoke out against the shutdown and the harm it caused those that rely upon public transit. One critic argued that buses could safely be kept running and that the decision "devastatingly impacts our marginalized populations, people [who] don't have access to vehicles and don't actually have the ability to access an Uber or a taxi, because it's very expensive" (quoted in Veneza, 2020). Several city councillors and other public officials also criticized the decision. The medical officer of health for Windsor-Essex, Dr. Wajid Ahmed, for example, made a public statement that included support for keeping transit service running:

> Under the right circumstances, such as employing appropriate environmental cleaning practices and managing the number and space of passengers, public transit can serve as an important means of transportation to access food, supplies, or go to work for essential workers who may not have a private vehicle or other options (quoted in Cross, 2020).

The decision to suspend public transit was reversed after a little more than four weeks following pressure from transit users, several city councillors, and other interested parties. Although short-lived, the decision did raise an important debate within the municipality about the importance of public transit and who should have decision-making authority in emergency situations. In the aftermath of the mayor's decision, the municipality's transit advisory committee recommended that public transit be declared an essential service. This was subsequently supported by a resolution passed by the Windsor's Environment, Transportation and Public Safety Committee and brought to a January 2021 council vote (Chen 2020; Cross, 2021a). The motion – which would have given the full city council control over transit decision-making in all circumstances – was supported by three city councillors but opposed by the mayor and the remaining seven councillors, thus leaving the existing decision-making power in place. In ordinary circumstances, the full council has authority over public transit decisions, while the mayor can have sole authority in times of emergency, such as the pandemic, and in these circumstances can make unilateral decisions.

The Windsor mayor's decision to suspend public transit service was not replicated in the other five municipalities examined here. There were also other differences in their immediate pandemic responses in part because each authority had to follow the guidelines established by their local medical officer of health and these guidelines were not always identical. In the case of

London, for example, the local medical officer did not mandate social distancing on buses making it the only one of the six authorities not to introduce social distancing rules at the outset of the pandemic (see Table 3). Nevertheless, there were many commonalities across the six authorities with respect to transit decision-making during the pandemic. They were, of course, responding to the same, largely unprecedented, crisis at the same time. It is not surprising therefore that they worked collaboratively and held regular meetings with the goal of determining best practices to keep their services running under the constraints presented by the pandemic. All the transit officials interviewed drew attention to the importance of these meetings in their planning. They also identified the regular meetings with the provincial government, and particularly the transportation minister, as well as with the federal government, as central influences on their decision-making.

All the interviewees also referenced the importance of transit users as an influence on decisions taken at the outset of the pandemic. One element of this was the input of advisory groups such as Windsor's transit advisory group mentioned above.[2] The interviewees also noted, however, that the pressures of responding to the pandemic meant that there was less regularized consultation with transit users and groups than occurs in normal times. They added that the primary way they heard from the public was through complaints registered with the authority or made directly to the bus operators. Instead, all six transit authorities found themselves focusing primarily on trying to respond quickly to changing health and safety guidance to protect their bus operators and to reassure passengers that public transit is safe.

[2] Windsor's transit advisory groups, as with other advisory groups in Windsor and indeed in other Ontario municipalities cannot make binding decisions. It is limited to providing recommendations and advice with respect to transit.

Transit users played a part in this process. In proposing and introducing health and safety measures, decision makers in all six transit systems responded to the complaints and concerns of transit users. Table 3 categorizes riders' concerns into four key categories: Service Coverage, Buses in Service, Social Distancing, and Violations of Health Guidelines.

Table 3: Riders' Concerns[3]

Municipality	Service coverage	Number of Buses in Service	Social Distancing	Violations of Health Guidelines
Brampton	Yes	Yes	Yes	Yes
Brantford	Yes	No	Yes	No
Burlington	Yes	No	No	Yes
Kingston	Yes	No	Yes	Yes
London	Yes	Yes	No	Yes
Windsor	Yes	Yes	Yes	Yes

The first two columns deal with complaints about transit coverage and frequency. As transit systems lowered their level of service due to financial constraints and safety concerns, riders that rely on public transit voiced their concerns that the service was insufficient. This is examined in more detail in the next section.

The third and fourth columns identify concerns relating to public health and thus the safety of public transit users in the context of the pandemic. The first of these, as identified by transit providers, are users' complaints that they are not able to socially distance while on a bus.

[3] These concerns were identified by our interviewees as well as through a survey of newspaper articles dealing with concerns about public transit.

As one Windsor interviewee identified, however, keeping two metres (approximately six feet) between riders would require a limit of only five riders on each bus compared to the current thirty-five, which is itself a significant reduction from pre-pandemic limits when riders were allowed to stand on buses.

The fourth concern focused on riders who do not wear a mask on buses or otherwise fail to follow health guidelines. Windsor is the only municipality of the six authorities to not allow exemptions to the mask mandate. As of May 14th, 2021, Transit Windsor has not allowed any rider to board without wearing a mask – this includes those that claim a medical exemption (Cross, 2021b). In the other authorities, it was determined that bus operators did not have the capacity to check if riders have a legitimate reason to board without a mask as per the Accessibility for Ontarians with Disabilities Act guidelines. As a result, in five of the six systems people are allowed to board and ride a bus without a mask. Brampton Transit, for example, made masks mandatory on July 2nd 2020, but identified that not everyone abides by the mask mandate once they are on the bus. London Transit similarly indicated that they have limited power to enforce compliance with the mask mandate or other COVID-19 restrictions, such as exiting through the rear door of buses (see also Trevithick, 2020). Overall, the six municipalities have introduced several health and safety measures in response to the pandemic and to users' concerns (see Table 4).

Table 4: Safety Measures

Municipality	Frequent Cleaning Practices	Rear door boarding	Limited Capacity	Plexiglass Barriers	Shutdown transit services
Brampton	Yes	Yes	Yes	Yes	No

Brantford	Yes	Yes	Yes	Yes	No
Burlington	Yes	Yes	Yes	Yes	No
Kingston	Yes	Yes	Yes	Yes[4]	No
London	Yes	Yes	Yes	Yes	No
Windsor	Yes	Yes	Yes	Yes	Yes—4-week period

Brampton was the first of the six municipalities to transition to rear-door boarding to protect their operators. At first, they only allowed 15 riders per bus but this was subsequently increased to 35. In a regular year, a bus capacity in Brampton is from 55 to 60 riders. Once masks became mandatory, Brampton Transit quickly returned to front-door boarding and was able to have people pay fares again.[5] An interviewee from Brampton noted that they were fortunate because they already had plexiglass barriers on each of their buses. Importantly, they have implemented 24-hour cleaning cycles. To make communications more accessible, they began publicising changes in public transit service through different cultural and ethnic television, radio, and other communication channels.

Other municipalities took similar initial approaches in their response to the pandemic. London, for example, also introduced nightly cleaning of all high-touch areas, and announcements were made each time the bus door opened reminding people of the safety protocols in place. London Transit also established a group of spare operators ready to provide

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[4] Kingston had plexiglass barriers on all buses by December 31st, 2021.

[5] In Brampton, as with the other municipalities, buses were not equipped to accept fare payment at their rear doors. As a result, rear-door boarding meant that Brampton allowed passengers to ride for free thus increasing the financial pressure on the transit authority. This was also the case in Kingston between April and August 2020 and Burlington between March and August 2020. Windsor was the last authority to return to front-door boarding – on October 19th 2000 – and thus the last to begin collecting fares again.

cover for those that call in sick. The Burlington Transit authority introduced new cleaning protocols whereby their spare operators not only covered illness but also began cleaning the interiors of buses. The interviewee from Burlington noted that this will continue in the post-pandemic era. And unlike other municipalities, Kingston Transit management did not deal with controlling public health messaging or enforcing the municipal mask mandate; instead, they allowed the municipality and its public health unit to take responsibility. This allowed them to focus their efforts on providing service rather than managing both service and health guidelines.

In the first months of the pandemic, then, transit decision-makers did seek to respond as best they could to users' complaints and concerns. Planning decisions were, however, affected by a whole range of factors and not only the views of disadvantaged groups and the goal of promoting transit equity. In the rapidly changing environment of the pandemic, decision-making could not always wait for consultations with all interested parties and the decisions taken did not always meet with the approval of groups seeking to promote transit equity.

4.3 Transit Decisions and Transit Equity During the Pandemic

The unexpected and unprecedented scale of the pandemic forced all transit authorities to make cuts to transit services. Windsor, as noted earlier, was the only city in Ontario to suspend completely its public transit network, although it reversed this decision after four weeks. When public transit resumed on May 4th, 2020, it was at a reduced 'Sunday level' of service before moving to a Saturday-level service in September 2020. Under a Saturday level of service, they provided service into the evenings but had less frequent buses during the day than in a non-pandemic year. The other jurisdictions also took decisions to eliminate some services and reduce the capacity of the buses that continued to run. Brampton, for example, decided to reduce services by maintaining its basic grid routes, the major routes that are most frequently used, while eliminating some of the less-used routes. London and Barrie also eliminated routes. Other transit networks maintained all routes but limited the frequency of bus service on these routes, as in the case of Windsor. This led one transit advocate to complain that the changes to transit schedules were unfair as riders still had to pay the same fare for “half of a service” once fare collection resumed (quoted in Aversa, 2020).

Transit authorities’ reduction in services, however necessary in response to the pandemic, inevitably negatively impacted the groups that have no or limited alternative transit options. New immigrants, seniors, individuals reliant on social services, and people in low-income jobs are among those who are most heavily reliant on public transit and who are most likely to live in areas underserved by public transit. They are also more likely to work in jobs – in food services sector for example – that cannot be undertaken remotely. Many could not work from home and nor could they decide to drive to work or to other essential locations. At one level, therefore, it can be simply stated that the first year or more of the pandemic had a negative impact on transit equity in the six municipalities.

It is, however, a mistake to be overly critical of the decisions made by transit authorities. This was an unprecedented crisis with a massive financial impact on public transit. Prior to the pandemic, for example, London Transit forecast that revenue from fares would cover approximately 46% of the authority’s budget in 2020. In the event, this revenue stream only covered 29.6%. In the case of Transit Windsor rider fares covered nearly half of transit’s expenditure prior to the pandemic. When the pandemic hit, this revenue stream initially disappeared completely and returned at only a fraction of the pre-pandemic levels. In February 2020 – one year into the pandemic – total operating revenues across the country were 68.7%

lower than pre-pandemic levels (Bogart, 2021). The combination of this reduced revenue and increased health and safety costs, made, and continues to make, funding a central worry for each transit authority. Even with additional financial support, cuts to transit services were unavoidable.

In the context of having to make cuts, transit decision-makers realized that these cuts negatively affected transit equity. Interviewees noted that the complaints regarding service coverage and frequency of service (as identified in Table 3) were most likely to come from the groups that rely most heavily on public transit. The interviewees all realised that transit is an essential service for many of their riders. One interviewee from Windsor, for example, commented on the frequency of complaints from seniors expressing their frustration with reduced Transit Windsor service. The same interviewee talked about the importance of public transit for people on the Ontario Disability Support Program (ODSP) or Ontario Works (OW):

> ODSP is about $733 a month and OW is about $1169 so by the time you cover your housing costs and your food that leaves very little for them to be able to use on taxi service to go to a pharmacy to get their medication, or get to a grocery store, or to work. So, as we've seen reintroduction of the buses it's still not at the level where it needs to be in order to be able to properly service the people who rely on it and disproportionately it [cuts in transit service] has had a negative impact on low-income earners and people with disabilities.

Several interviewees noted that cuts to service routes and service frequency negatively affected front-line workers in low-paid jobs. One interviewee, for example, mentioned a story of a frequent low-income rider who had to resort to using taxi services to get to and from work, which left them in a financial crisis. Interviewees talked about the importance of maintaining transit service for low-income groups but also about the importance of providing a service that will attract all socio-economic classes noting that a successful public transit network requires a broad range of riders.[6]

Transit officials in all six authorities were aware of the threat that the pandemic poses to a goal of increased transit equity and sought to ameliorate its impact on marginalized communities. They recognized the reality of the point made by Marco D'Angelo, president of the Canadian Urban Transit Association: "Workers, seniors, students, and people living with disabilities rely on public transit every day. It's critical it remains a safe and sustainable option for travel both during and after COVID-19" (quoted in "Ontario Increases Support for Transit Across the Province," 2021). Indeed, transit officials made this part of their campaign to secure emergency funding from their municipalities and support from senior governments. In the words of one official:

> If all of a sudden the province and the feds don't give us that $20 or 25 million…and they said well transit you're on your own you gotta do something whatever actions we're going to take to come up with $20 to 25,000,000 million are going to result in for sure way less service on the street. Those people going to those essential jobs being crowded on even more crowded buses in the height of a worldwide pandemic so we're hopeful our budget hole will be filled….

In each case, transit authorities did receive emergency funding to keep the service running. As Table 5 identifies, in 2020 the municipal governments did make up the shortfall from the loss of fare revenues and each transit authority's operating budget remained similar to, or was higher than, the 2019 budget despite the drop in fare revenue.

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[6] This is mentioned by a wide range of transit planning experts (see Schwartz, 2015).

Table 5: Transit Operating Budgets

Function	2019 Budget Year	Dollar Change from 2018 (%)	2020 Budget Year	Dollar Change from 2019 (%)
Brampton Transit	$68,925,000	$5,654,000 (8.2%)	$72,586,000	3,661,000 (+5%)
Brantford Transit	$6,314,812	$116,531 (1.8%)	$6,311,629	$3,183 (-0.05%)
Burlington Transit	$13,072,000	$2,031,000 (15.5%)	$14,305,000	1,233,000 (+9.4%)
Kingston Transit	$17,146,360	$971,640 (5.7%)	$17,542,850	396,490 (+2.3%)
London Transit Commission	$32,831,000	1,301,400 (4%)	$38,089,000	5,258,000 (+13.8%)
Transit Windsor	$36,061,034	$2,129,926 (6.3)	$36,294,433	$233,399 (+0.6)

The transit authorities also received large injections of funds from the provincial and federal governments. In July 2020, the federal Liberal government announced the Safe Restart Agreement, allocating $2 billion in federal funding to municipalities across Ontario, with $1 billion going to transit systems (Infrastructure Canada, 2020). The Ontario government also contributed $2.2 billion to the Safe Restart Agreement, with $1 billion going towards transit systems (Ford, 2020; "Ontario Increases Support for Transit Across the Province," 2021). Table 6 shows that all six transit authorities received substantial injections of the Safe Restart Funding allocated to public transit. Phase One was an allocated amount to each transit authority without applications and based on transit ridership. Phase Two of funding was distributed based on applications to cover expenses incurred as a result of the pandemic between October 1st 2020 and March 31st 2021. Phase Three funding was available for the period between April 1st and

December 31st 2021, which as well as covering pandemic expenses could be used to finance experiments with on-demand transit and regional transit initiatives. Interviewees stressed the importance of Safe Restart Funding in covering lost revenue as well as the health and safety costs that they did not have in their budget pre-pandemic.

Table 6: Senior Government Support for Transit

Municipality	**Safe Restart Agreement (Phase 1)** Apr 1 – Sep 30 2020	**Safe Restart Agreement (Phase 2)** Oct 1 2020 – Mar 31 2021	**Safe Restart Agreement (Phase 3)** Apr 1 – Dec 31 2021
Brampton	$24 million	$30.1 million	$23.5 million
Brantford	$1.2 million	$750,000	$1.2 million
Burlington	$2.6 million	$1.04 million	$1.5 million
Kingston	$5.32 million	$6.7 million	$5.19 million
London	$18.5 million	$23.18 million	$18.15 million
Windsor	$6.4 million	$7.945 million	$6.1 million

An argument made by many transit authorities and advocates of public transit is that the key to keeping public transit viable in the future is funding to fill in the gaps created due to COVID-19 as well as to also cover costs to enhance service and build trust in public transit within their communities. The City of Brampton asserted that "ongoing [senior government] funding" was needed "to ensure essential workers riding public transit have the means to get to and from work and for others to take essential trips" (2021). The Canadian Urban Transit Association similarly points to the importance of ongoing government support for public transit:

> Governments recognized transit's essential role in daily life by including it in the safe restart agreement. Transit systems and their riders are grateful, but the need isn't going away. Many of the millions of people who take transit daily have no other mobility options. If service is reduced as funding ends, people who live too far from work or school to walk or cycle but whose income is too low to drive will endure long waits and crowded vehicles. Governments can't let that happen, but time is running out (CUTA, 2021; see also D'Angelo, 2020).

All the interviewees in this study directly or indirectly stressed that the importance of public transit to marginalized individuals was central to their requests for financial support from senior governments. Senior government funding was seen as vital to addressing the challenges created by COVID-19 and furthermore that this support was essential to the promotion of transit equity. One interviewee noted that this was a big "a-ha" moment in which the federal and provincial governments recognized public transit as an essential service to keep communities moving.

There is some evidence to suggest that this campaign was effective. In providing the Safe Restart Agreement Funding, the Ontario government stated that the funding can go towards new initiatives that will make transit services *both more affordable and accessible* ("Windsor to receive $6 Million in transit funding", 2021, emphasis added).

The federal government has also played a key role in supporting public transit during the pandemic. As noted above, this support came in the shape of funding under the Safe Restart Agreement. In February 2021, for example, Prime Minister Trudeau announced additional funding beginning in 2026 towards public transit. The funding is estimated to be more than $13 billion and to be provided to 1,300 public transit projects across the country (Prime Minister of

Canada, 2021). The announcement of this funding commitment also drew attention to the importance of support for transit equity. An Infrastructure Canada backgrounder noted that:

> Public transit, rural transit solutions and active transportation infrastructure continue to provide reliable, fast, affordable and clean ways for people to get around. *These benefits are felt the most by disadvantaged groups for whom car travel isn't accessible*. Essential workers have relied on buses, subways, ride-sharing programs and pathways to get to where they are needed in grocery stores, hospitals and care facilities (Infrastructure Canada, 2021, emphasis added).

It is, however, necessary not to overstate the extent to which the federal and provincial governments have emphasized transit equity in their support for public transit during the pandemic. While transit authority representatives and interest groups talked about the importance of transit equity, no press release or minister's speech explicitly used the term 'transit equity'. The primary emphasis in the government announcements was on providing funding to keep existing transit services functioning through the pandemic. The Ontario Minister of Transportation Caroline Mulroney, for example, asserted that the government was responding to municipalities' "need for more support as COVID-19 continues to result in lost revenue and additional costs for transit systems…transit operators have done an incredible job keeping transit operating during this challenging time, and this support will help ensure there is reliable transportation for people to get to work or pick up essential items" (Ontario Government, 2021). The federal government also pointed to the importance of the Safe Restart Funding in allowing authorities to keep existing public transit operating.

Only part of the Safe Restart Agreement funding for transit has been directly aimed at potentially advancing the goal of transit equity beyond the important goal of keeping existing services running. Municipalities in Ontario have been able to apply for funding under Phase 3 to support “new initiatives that make it easier and more affordable to travel between different transit systems, like adding capacity for on-demand micro-transit” (Ontario Government, 2021). Indeed, this latter initiative provides an example of the way in which transit equity was at least considered by transit authorities during the first year of the pandemic. Transit authorities experimented with innovative approaches to providing transit service in under-serviced areas. Transit Windsor, for example, sought to address problems arising from reduced bus frequency and limits on bus capacity by implementing real-time service provision, a form of public transit on demand. They placed twenty buses strategically around the city and a driver of another full bus could call and let them know that their bus had reached its pandemic-reduced capacity. One of the twenty buses closest to said stop would then drive there. A proposed experiment with on-demand service had been included in Windsor Transit’s transit masterplan created in 2019. The pandemic served to speed up the examination of an official on-demand transit service in Windsor in which riders can use an app and request a bus to pick them up. This would be used in areas of low ridership to both save money and simultaneously be more accessible to riders when they need it. The city of London, Ontario is also looking to implement this for industrial areas in the city. Brampton Transit provides another example of a network responding to problems identified by transit users and at least attempting to make decisions with the goal of transit equity in mind. Decision-makers in Brampton knew that their riders did not all have English as a first language, and as such, they ensured that they made all communications accessible to other cultural and ethnic communities to keep riders informed of transit changes as well as to receive feedback from these communities.

5. Conclusions

The problems identified in transit equity literature are real issues that have been intensified by the COVID-19 pandemic. Issues regarding access to public transit were exacerbated by pandemic-induced cuts to service. As a result of these cuts, many who rely on public transit to get to work were left with fewer, less reliable or far less affordable, means to get to work.

The pandemic has affected all six transit authorities examined here similarly. They have all faced problems linked to declining ridership, loss of fare revenue, increased costs linked to cleaning, and all have had to make decisions about service provision in the face of declining use. In each case, the pandemic has been felt differently within municipalities. Each municipality has different groups that rely upon public transit to get to necessary locations, and it is evident that changes made to transit are not uniformly felt by all citizens. Changes made to public transit services, particularly those at the start of the pandemic, disproportionately impacted low-income citizens and those already marginalized.

A key, positive, conclusion from this study is that transit policy-makers are aware of the importance of transit equity, the negative impact of the pandemic on transit equity, and have tried to take steps to limit this impact. Although the necessity of responding to the pandemic quickly limited the role of consultation with transit users in the initial weeks and months of the crisis, each interviewee demonstrated awareness of transit equity and a professed desire to promote transit equity. In addition, the pandemic has also prompted transit decision-makers to look at innovative service options and technological advances to address gaps in transit equity. The idea of "transit on-demand" is being explored as a solution to ensuring populations are not left without reliable transit service as well as to encourage people who do not utilize public transit to begin to do so. It can also potentially increase the grid that transit systems utilize and thus, reach a bigger population of riders that may usually not have access to transit — such as those in rural communities.

There is, however, a limit to what transit authorities can do on their own to support public transit through the pandemic. While they can recommend technological innovations, on-demand services, and try to maintain services in communities that depend on public transit the most, none of it can be made possible without increased support and funding in the future. Every interviewee stressed the importance of funding to increase service and be better prepared in the future. The pandemic has increased the costs of providing public transit while it has dramatically reduced the fare revenues collected from public transit users. As a result, all six transit authorities have had to rely on increased municipal funding combined with subsidies from senior levels of government to continue their operations. The evidence indicates that, thus far, the Ontario and Canadian governments understand that the provision of public transit is important and does require financial support. Both provincial and federal governments have provided massive funding injections to municipalities and political leaders up to and including Prime Minister Trudeau have emphasized the importance of public transit on many different aspects of society and have been working to keep it running.

Senior governments will have to continue the financial support in order to support public transit in the future. The COVID-19 pandemic has had a huge, and negative impact on public transit. It is likely that the challenges posed by the pandemic will have a lingering effect on the future of public transit provision. Transit authorities cannot be left alone to deal with the problems of providing public transit in what remains of the COVID era. Senior levels of government must provide ongoing financial support if public transit is to survive.

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